StudySapiens Board MCQs

Chapter 6 — Bioenergetics

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The universal energy currency of cells is:

(a) DNA
(b) Glucose
(c) NADH
(d) ATP

2. ATP is composed of adenine, ribose and:

(a) One phosphate group
(b) Four phosphate groups
(c) Two phosphate groups
(d) Three phosphate groups

3. When ATP is hydrolysed to ADP and Pi, it releases about:

(a) 100 kcal/mole
(b) 7.3 kcal/mole
(c) 20 kcal/mole
(d) 1 kcal/mole

4. The ATP-ADP cycle links:

(a) DNA and RNA only
(b) Photosynthesis and transpiration only
(c) Energy-releasing and energy-requiring reactions
(d) Protein and lipid only

5. Photosynthesis converts light energy into:

(a) Heat only
(b) Electrical energy only
(c) Chemical energy
(d) Mechanical energy only

6. The overall products of photosynthesis are:

(a) Alcohol and CO2
(b) CO2 and water
(c) Glucose and oxygen
(d) Lactic acid and ATP

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7. The oxygen released during photosynthesis comes from:

(a) Carbon dioxide
(b) Water
(c) ATP
(d) Glucose

8. Van Neil's hypothesis proposed that plants split:

(a) CO_2 to obtain oxygen
(b) ATP to obtain phosphate
(c) Glucose to obtain water
(d) Water to obtain hydrogen

9. The action spectrum of photosynthesis shows maximum activity in:

(a) Green light only
(b) Blue and red light
(c) Far-red light only
(d) Yellow light only

10. The first action spectrum was obtained using:

(a) Elodea only
(b) Hydrilla
(c) Chlorella
(d) Spirogyra

11. Carbon dioxide enters leaves mainly through:

(a) Cuticle
(b) Stomata
(c) Xylem
(d) Phloem

12. The primary photosynthetic pigment is:

(a) Phycoerythrin
(b) Carotene
(c) Chlorophyll a
(d) Xanthophyll

13. Accessory pigments help photosynthesis by:

(a) Forming glucose directly
(b) Absorbing additional wavelengths and transferring energy to chlorophyll a
(c) Producing CO_2
(d) Replacing chlorophyll a

14. Photosystems are located in the:

(a) Thylakoid membranes
(b) Matrix
(c) Outer chloroplast membrane
(d) Stroma

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15. Light-dependent reactions produce:

(a) Pyruvic acid only
(b) Glucose only
(c) ATP and NADPH
(d) CO_2 and water

16. Non-cyclic photophosphorylation produces:

(a) ATP only
(b) Glucose only
(c) NADPH only
(d) ATP, NADPH and oxygen

17. Cyclic photophosphorylation produces:

(a) ATP and NADPH
(b) NADPH and oxygen
(c) Glucose and oxygen
(d) ATP only

18. Photolysis of water occurs in:

(a) Photosystem I only
(b) Calvin cycle
(c) Krebs cycle
(d) Photosystem II

19. The final electron acceptor in non-cyclic photophosphorylation is:

(a) ATP
(b) CO_2
(c) $NADP^+$
(d) Oxygen

20. Chemiosmosis in chloroplasts involves movement of H^+ through:

(a) DNA polymerase
(b) ATP synthase
(c) Cytochrome oxidase
(d) Rubisco

21. The Calvin cycle occurs in the:

(a) Stroma
(b) Cytoplasm
(c) Matrix
(d) Thylakoid lumen

22. The enzyme that fixes CO_2 in the Calvin cycle is:

(a) Pepsin
(b) Rubisco
(c) ATP synthase
(d) Hexokinase

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23. The CO_2 acceptor in the Calvin cycle is:

(a) Glucose
(b) G3P
(c) RuBP
(d) PGA

24. The first stable product of carbon fixation in C3 plants is:

(a) Malic acid
(b) PGA
(c) Oxaloacetic acid
(d) Glucose

25. The Calvin cycle uses ATP and NADPH to reduce:

(a) $NADP^+$ to NAD^+
(b) CO_2 to sugar
(c) Glucose to pyruvate
(d) Water to oxygen

26. For every three CO_2 molecules fixed, the Calvin cycle can produce one molecule of:

(a) Glucose
(b) G3P
(c) ATP
(d) RuBP

27. To produce one glucose molecule, the Calvin cycle must fix:

(a) Three CO_2 molecules
(b) Twelve CO_2 molecules
(c) Six CO_2 molecules
(d) One CO_2 molecule

28. Cellular respiration is the process of:

(a) Producing oxygen
(b) Releasing energy from organic molecules
(c) Storing light energy
(d) Fixing CO_2

29. The common respiratory fuel is:

(a) Glucose
(b) Oxygen
(c) CO_2
(d) Water

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30. Glycolysis occurs in the:

(a) Nucleus
(b) Mitochondrial matrix
(c) Thylakoid
(d) Cytoplasm

31. Glycolysis breaks glucose into:

(a) Six CO_2 molecules
(b) Two pyruvic acid molecules
(c) Two acetyl CoA molecules
(d) One lactic acid molecule only

32. The net ATP yield of glycolysis is:

(a) 2 ATP
(b) 36 ATP
(c) 4 ATP
(d) 0 ATP

33. Glycolysis also produces:

(a) Oxygen
(b) NADH
(c) $FADH_2$ only
(d) Glucose

34. Anaerobic respiration in yeast produces:

(a) Alcohol and CO_2
(b) Glucose only
(c) Oxygen and water
(d) Lactic acid only

35. Anaerobic respiration in human muscle cells produces:

(a) Alcohol
(b) Lactic acid
(c) Oxygen
(d) Glucose

36. Anaerobic respiration yields only:

(a) 2 ATP per glucose
(b) 4 ATP per glucose
(c) 36 ATP per glucose
(d) 38 ATP per glucose

37. Pyruvic acid is converted to acetyl CoA in the:

(a) Cytoplasm
(b) Mitochondrion
(c) Chloroplast
(d) Nucleus

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38. During pyruvic acid oxidation, one molecule of CO_2 is released and:

(a) Oxygen is produced
(b) NADH is formed
(c) Glucose is formed
(d) $FADH_2$ is formed only

39. The Krebs cycle occurs in the:

(a) Thylakoid
(b) Inner membrane only
(c) Mitochondrial matrix
(d) Cytoplasm

40. Acetyl CoA combines with oxaloacetic acid to form:

(a) Citric acid
(b) Malic acid only
(c) Lactic acid
(d) Pyruvic acid

41. For each glucose molecule, the Krebs cycle turns:

(a) Once
(b) Three times
(c) Twice
(d) Six times

42. The Krebs cycle produces per glucose:

(a) 36 ATP only
(b) 2 ATP, 6 NADH and 2 $FADH_2$
(c) 2 ATP only
(d) No NADH

43. The electron transport chain is located in the:

(a) Inner mitochondrial membrane
(b) Outer mitochondrial membrane
(c) Cytoplasm
(d) Matrix only

44. The final electron acceptor in aerobic respiration is:

(a) FAD
(b) NAD^+
(c) Oxygen
(d) CO_2

45. Chemiosmosis in mitochondria uses a proton gradient across the:

(a) Nuclear envelope
(b) Plasma membrane only
(c) Inner mitochondrial membrane
(d) Outer membrane

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46. According to the textbook, one NADH from the Krebs cycle yields:

(a) 2 ATP
(b) 3 ATP
(c) 4 ATP
(d) 1 ATP

47. According to the textbook, one FADH2 yields:

(a) 3 ATP
(b) 2 ATP
(c) 4 ATP
(d) 1 ATP

48. The textbook net yield of aerobic respiration per glucose is:

(a) 18 ATP
(b) 4 ATP
(c) 2 ATP
(d) 36 ATP

49. Aerobic respiration is about how many times more efficient than anaerobic respiration in the textbook?

(a) 10 times
(b) 18 times
(c) 5 times
(d) 2 times

50. Glycolysis NADH yields fewer ATP in the textbook balance sheet because:

(a) It is used in photosynthesis
(b) It becomes FADH2
(c) Transport into mitochondria costs energy
(d) It is not produced

51. Fats are excellent respiratory fuels because they contain many:

(a) C-H bonds
(b) Phosphodiester bonds
(c) Peptide bonds
(d) Glycosidic bonds only

52. Glycerol enters respiration after conversion to:

(a) Glyceraldehyde 3-phosphate
(b) Acetyl CoA only
(c) Oxaloacetic acid only
(d) Citric acid

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53. Fatty acids enter respiration mainly after conversion to:

(a) Pyruvic acid only
(b) Acetyl CoA
(c) Glucose
(d) Lactic acid

54. Amino acids enter respiration after:

(a) Translation
(b) Deamination
(c) Photosynthesis
(d) Transcription

55. Photorespiration occurs when rubisco fixes:

(a) Nitrogen instead of CO_2
(b) CO_2 instead of O_2
(c) Water instead of CO_2
(d) Oxygen instead of CO_2

56. Photorespiration results in:

(a) Glucose synthesis
(b) Oxygen production
(c) Net ATP production
(d) Loss of fixed carbon as CO_2

57. Photorespiration is especially problematic in:

(a) Aquatic algae only
(b) CAM plants only
(c) C4 plants only
(d) C3 plants at high temperature

58. C4 plants initially fix CO_2 into:

(a) RuBP only
(b) Oxaloacetic acid
(c) PGA only
(d) Glucose

59. In C4 plants, the Calvin cycle occurs mainly in:

(a) Guard cells
(b) Bundle-sheath cells
(c) Mesophyll cells only
(d) Epidermal cells

60. C4 plants include:

(a) Corn and sugarcane
(b) Wheat and rice only
(c) Pine only
(d) Cacti only

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61. CAM plants open their stomata:

(a) Never
(b) During the day only
(c) At night
(d) Only in rain

62. CAM plants store CO_2 at night as:

(a) Starch only
(b) Glucose only
(c) Oxygen
(d) Organic acids

63. The compensation point is when:

(a) Only transpiration occurs
(b) Photosynthesis and respiration rates are equal
(c) Photosynthesis stops
(d) Respiration stops

64. Which process produces oxygen?

(a) Glycolysis
(b) Fermentation
(c) Light-dependent reactions of photosynthesis
(d) Krebs cycle

65. Which process consumes oxygen?

(a) Calvin cycle
(b) Aerobic respiration
(c) Photosynthesis
(d) Glycolysis only

66. NADPH is mainly used in:

(a) The Calvin cycle
(b) Glycolysis
(c) Krebs cycle
(d) Electron transport chain only

67. The proton gradient in chloroplasts is formed across the:

(a) Inner chloroplast envelope only
(b) Thylakoid membrane
(c) Outer membrane
(d) Stroma lamellae only

68. The enzyme ATP synthase is involved in:

(a) DNA replication
(b) Carbon fixation only
(c) Chemiosmotic ATP synthesis
(d) Protein digestion

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69. Which of the following is a product of the light reactions?

(a) Pyruvate
(b) Lactic acid
(c) Glucose
(d) NADPH

70. Which of the following is NOT part of aerobic respiration?

(a) Krebs cycle
(b) Electron transport chain
(c) Glycolysis
(d) Calvin cycle

71. The link reaction converts pyruvate to:

(a) Glucose
(b) Acetyl CoA
(c) Ethanol
(d) Lactate

72. The majority of ATP in aerobic respiration is produced by:

(a) Oxidative phosphorylation
(b) Glycolysis only
(c) Substrate-level phosphorylation only
(d) Fermentation

73. Substrate-level phosphorylation occurs in:

(a) Glycolysis and Krebs cycle
(b) Only ETC
(c) Only light reactions
(d) Only Calvin cycle

74. In the absence of oxygen, the electron transport chain stops because:

(a) ATP is absent
(b) NADH is absent
(c) There is no final electron acceptor
(d) Glucose is absent

75. The main purpose of fermentation is to:

(a) Produce oxygen
(b) Regenerate NAD+ for glycolysis
(c) Fix CO2
(d) Produce large amounts of ATP

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76. Which molecule carries electrons from glycolysis and Krebs cycle to the ETC?

(a) ATP only
(b) NADH and FADH2
(c) Glucose only
(d) CO2 only

77. The textbook value of ATP from one glucose in aerobic respiration is:

(a) 36
(b) 32
(c) 30
(d) 38

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ANSWER KEY WITH BRIEF REASONS

1. (d) ATP — ATP stores and transfers usable energy.
2. (d) Three phosphate groups — ATP = adenosine triphosphate.
3. (b) 7.3 kcal/mole — ATP hydrolysis releases usable energy.
4. (c) Energy-releasing and energy-requiring reactions — ATP transfers energy between processes.
5. (c) Chemical energy — Light energy is stored in organic molecules.
6. (c) Glucose and oxygen — Photosynthesis produces sugar and O2.
7. (b) Water — Photolysis of water releases oxygen.
8. (d) Water to obtain hydrogen — Water is the source of reducing power and O2.
9. (b) Blue and red light — Chlorophyll absorbs blue/red most effectively.
10. (d) Spirogyra — Engelmann used Spirogyra.
11. (b) Stomata — Stomata allow gas exchange.
12. (c) Chlorophyll a — Chlorophyll a is the main reaction-centre pigment.
13. (b) Absorbing additional wavelengths and transferring energy to chlorophyll a — They broaden light absorption.
14. (a) Thylakoid membranes — Light reactions occur in thylakoids.
15. (c) ATP and NADPH — They convert light energy into chemical carriers.
16. (d) ATP, NADPH and oxygen — Electrons flow from water to NADP+.
17. (d) ATP only — Cyclic flow makes ATP without NADPH/O2.
18. (d) Photosystem II — PSII splits water.
19. (c) NADP+ — NADP+ is reduced to NADPH.
20. (b) ATP synthase — Proton flow drives ATP synthesis.
21. (a) Stroma — Carbon fixation occurs in stroma.
22. (b) Rubisco — Rubisco catalyses CO2 + RuBP.
23. (c) RuBP — RuBP combines with CO2.
40. (a) Citric acid — Citrate synthase forms citrate.
41. (c) Twice — Two pyruvates enter per glucose.
42. (b) 2 ATP, 6 NADH and 2 FADH2 — These carriers feed the ETC.
43. (a) Inner mitochondrial membrane — ETC complexes are in the inner membrane.
44. (c) Oxygen — Oxygen forms water at the end of the chain.
45. (c) Inner mitochondrial membrane — H+ flow through ATP synthase makes ATP.
46. (b) 3 ATP — Classical textbook yield is 3 ATP per NADH.
47. (b) 2 ATP — Classical textbook yield is 2 ATP per FADH2.
48. (d) 36 ATP — The book uses 36 ATP net.
49. (b) 18 times — 36 ATP vs 2 ATP.
50. (c) Transport into mitochondria costs energy — Cytoplasmic NADH shuttling costs ATP.
51. (a) C-H bonds — C-H bonds release much energy on oxidation.
52. (a) Glyceraldehyde 3-phosphate — Glycerol becomes a glycolysis intermediate.
53. (b) Acetyl CoA — Beta-oxidation produces acetyl CoA.
54. (b) Deamination — The amino group is removed first.
55. (d) Oxygen instead of CO2 — Rubisco has oxygenase activity.
56. (d) Loss of fixed carbon as CO2 — It wastes fixed carbon.
57. (d) C3 plants at high temperature — Heat increases rubisco oxygenase activity.
58. (b) Oxaloacetic acid — PEP carboxylase forms a 4-carbon acid.
59. (b) Bundle-sheath cells — CO2 is concentrated in bundle-sheath cells.
60. (a) Corn and sugarcane — Corn/sugarcane are typical C4 plants.
61. (c) At night — Night opening reduces water loss.
62. (d) Organic acids — CO2 is fixed into acids for daytime use.

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24. (b) PGA — 3-phosphoglycerate is formed first.

25. (b) CO_2 to sugar — Carbon reduction requires energy/reducing power.

26. (b) G3P — One G3P can leave the cycle.

27. (c) Six CO_2 molecules — Glucose has six carbons.

28. (b) Releasing energy from organic molecules — Respiration oxidizes fuel to make ATP.

29. (a) Glucose — Glucose is the usual substrate.

30. (d) Cytoplasm — Glycolysis is cytosolic.

31. (b) Two pyruvic acid molecules — Glucose is split into two 3-carbon pyruvates.

32. (a) 2 ATP — 2 ATP are used and 4 produced, net 2.

33. (b) NADH — NAD+ is reduced to NADH.

34. (a) Alcohol and CO_2 — Alcoholic fermentation occurs in yeast.

35. (b) Lactic acid — Lactic acid fermentation occurs during oxygen deficit.

36. (a) 2 ATP per glucose — Fermentation yields little ATP.

37. (b) Mitochondrion — Pyruvate oxidation occurs in mitochondria.

38. (b) NADH is formed — Pyruvate loses CO_2 and reduces NAD+.

39. (c) Mitochondrial matrix — Matrix enzymes catalyse the cycle.

63. (b) Photosynthesis and respiration rates are equal — Net gas exchange is zero.

64. (c) Light-dependent reactions of photosynthesis — Photolysis releases O_2.

65. (b) Aerobic respiration — Oxygen is the final electron acceptor.

66. (a) The Calvin cycle — NADPH provides reducing power for carbon fixation.

67. (b) Thylakoid membrane — H+ accumulates in the thylakoid space.

68. (c) Chemiosmotic ATP synthesis — Proton flow drives ATP formation.

69. (d) NADPH — Light reactions make ATP/NADPH.

70. (d) Calvin cycle — Calvin cycle is photosynthetic.

71. (b) Acetyl CoA — Pyruvate oxidation forms acetyl CoA.

72. (a) Oxidative phosphorylation — ETC/chemiosmosis makes most ATP.

73. (a) Glycolysis and Krebs cycle — Direct phosphate transfer makes ATP.

74. (c) There is no final electron acceptor — Oxygen is required to accept electrons.

75. (b) Regenerate NAD+ for glycolysis — NAD+ regeneration allows glycolysis to continue.

76. (b) NADH and $FADH_2$ — Reduced coenzymes donate electrons.

77. (a) 36 — PECTAA textbook uses 36 ATP.

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